ACLS Track 1

Introduction to Bioinformatics Centre Projects

Tugce Bilgin Sonay, ZHAW

meet Tugce

UZH



ZHAW



UniL



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- Tugce – say it like 2Che
- lecturer at ZHAW
- PhD and postdoctoral research in Switzerland
- lecturer at Columbia University, NYC
  - also taught at Brooklyn College of CUNY and Pratt Institute of Art and Design
- teach science (data analysis, biostatistics, bioinformatics, physics), art (creative coding at museums) and yoga for refugees
- supervise student projects in bioinformatics
- teach these bioinformatics workshops since 2016

Meet us

Research Centre for Bioinformatics

We develop practical solutions at the interface of biology, medicine and computational sciences.



The Research Centre for Bioinformatics focuses on the theoretical and computational aspects of modelling the molecular biology processes, genome evolution and adaptive change, as well as biomedical data representation and integration. The goal is to bring basic research and new bioinformatics methods to real-world applications, ranging, for example, from biotechnology and forensics to biomedical research and environmental applications. The research area is represented by the several research groups, each focusing on certain methods or application domains.

Meet us

Computational Genomics Lab:

Maria Anisimova

Olesia Kondrateva

Feifei Xia

Tugce Bilgin Sonay









Research Centre for Bioinformatics at ZHAW

Computational Genomics

The research group develops computational methods for comparative and evolutionary genomics, including modelling of stochastic processes in molecular evolution. Many research projects focus on the analysis of protein-coding genes and gene families, selection, adaptation, phylodynamics and evolution, including host-pathogen interactions; applications in medical genomics, epidemiology, metagenomics and forensics. Our research includes studies of genomic repeat sequences and indel evolution with applications in cancer genomics and biotechnology, as well as studies of dynamics and evolution of viruses and other pathogens.

Group leader: Prof. Dr. Maria Anisimova

https://github.com/acg-team/Bioinfo4B



Research Centre for Bioinformatics at ZHAW

Biomedical String Analysis

The research group is specialized in the analysis of strings (i.e. finite sequences of symbols). The research projects and applications focus on genomic data and biomedical natural language. The group develops new computational science methods and applies existing methods. This includes: mathematical modeling, computational statistics, algorithm design, discrete mathematics, machine and deep learning, natural language processing, semantic web technologies.

Group leader: Dr. Manuel Gil | Learn more about the research group Biomedical String Analysis

Applied Mathematical Biology

The research group develops and applies mathematical models and methods to address open research questions in biology. Many methods use standard calculus, differential equations, machine learning and dynamical systems theory to describe and predict biological phenomena, such as for example, the relationship between codon bias and gene expression via the concept of translational efficiency, applied to codon optimization problems. Further interests lie in the exploration of cancer-immune system interactions and their predictive power for cancer immunotherapies as well as the population genetics of the early infection-phase of partially-recombining viruses.

Group leader: Dr. Victor Garcia